

GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 16



By- Amulya Ratan Sir

Recap of Previous Lecture



Topic

Data Interpretation



Topics to be Covered



Topic-1

Interest



[MCQ]



#Q. By approximately what percent, the strength of school increased or decreased from 2007 to 2008?

Assignment

A

1.2%

B

1.7%

C

2.1%

D

2.4%

$$\frac{50}{2950} \times 100$$

$$= \frac{500}{295}$$

$$= 1.7\%$$

(3000)
2005

Strength of school in 2005 was 3000



[MCQ]



#Q. How many students were studying in the school during 2009?

Assignment

Strength of school in 2005
was 3000

- A** 2950
- B** 3000
- C** 3100
- D** 3150



[MCQ]



2008 (4)
2011

#Q. The number of students studying in school in 2008 was what percent of the number of students studying in the school in 2011?

Assignment

Strength of school in 2005 was 3000

$$\frac{1500}{3000} \times 100 = 50\%$$
$$\frac{450}{500} \times 100 = 90\%$$
$$50\% + 43.75\% = 93.75\%$$

A

92.13%

B

93.75%

C

97.25%

D

96.88%



[MCQ]



#Q. Who among the following employees earns maximum bonus in comparison to his total income?

Assignment

A

P

B

Q

C

R

D

S

$$P \Rightarrow \frac{80}{900} \times 100 = 8.8\%$$

$$Q \Rightarrow 8\%$$

$$R \Rightarrow 10\%$$

$$S \Rightarrow \frac{80}{7} = 11.43\%$$

$$T \Rightarrow 10\%$$

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]



#Q. The income from overtime is what percent of income from arrears in the case of employee P?

Assignment

- A** 90
- B** 80
- C** 40
- D** 75

$$\frac{180}{200} \times 100 = 90\%$$

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]



#Q. How many employee have their salary more than 3 times the income from bonus?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A

One

B

Five

C

Three

D

None

[MCQ]



#Q. Who among the following employees has maximum percentage of his salary out of the total income?

Assignment

A

T

B

P

C

S

D

R

$$P \Rightarrow \frac{400}{9} \% = 44.4\%$$

$$Q \Rightarrow 40\%$$

$$R \Rightarrow \frac{700}{15} \% = 46.6\%$$

$$S \Rightarrow \frac{380}{7} \% = 54.3\%$$

$$T \Rightarrow 40\%$$

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]

Arrears
Sal



#Q. Who among the following employees has minimum ratio of income from arrears to the income from salary?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

$P \Rightarrow \frac{200}{400} = \frac{1}{2} = 0.5$

$Q \Rightarrow \frac{180}{200} = \frac{9}{10} = 0.9$

$R \Rightarrow \frac{4}{7} = 0.57$

$S \Rightarrow \frac{14}{30} = \frac{7}{15} = 0.46$

$T \Rightarrow \frac{250}{400} = \frac{5}{8} = 0.625$

A

P

B

Q

C

T

D

S

INTEREST

(P) Principal

$$(A) \text{ Amount} = \underline{\underline{P}} + \underline{\underline{I}}$$

C.I.

121

110

S.I.

1204

Simple Interest

$$I = \frac{PRT}{100}$$

+P +P
 A = P \left(1 + \frac{TR}{100} \right)

Handwritten annotations:
 - "Time" is circled in blue with an arrow pointing to 'T'.
 - "Rate" is written in orange with an arrow pointing to 'R'.
 - "Principal" is written in orange with an arrow pointing to 'P'.
 - The '100' in the denominator of the first formula is underlined in orange.

Compound Interest

$$T = 3 \text{ yrs}$$

Annually $\Rightarrow n=3$

Half $\Rightarrow n=6$

Quarterly $\Rightarrow n=12$

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$(-P)$
 $(-P)$

$$I = P \left[\left(1 + \frac{r}{100} \right)^n - 1 \right]$$

Note:

3,00,000 ✓

✓ EMI

✓ 10k ✓

✓
15 lacs

3 years

3,60,000

✓

1st

2nd

3rd

Further formula:

Diffrence
between SI & CI
for 2 yrs

$$C.I - S.I$$

$$= P \left(\frac{R}{100} \right)^2$$

$$P \left(1 + \frac{R}{100} \right)^2 - P \left(1 + \frac{2R}{100} \right)$$

$$P \left[1 + \left(\frac{R}{100} \right)^2 + \frac{2R}{100} - 1 - \frac{2R}{100} \right]$$

[MCQ]



#Q. An investment triples itself in 10 years. In how many years it would be nine times itself?

$$A = P \left(1 + \frac{TR}{100} \right)$$

$$A = 3P \rightarrow 10 \text{ yrs}$$

$$I = 2P$$

(x4)

(x4)

$$I = 8P \rightarrow 40 \text{ yrs}$$

$$A = 9P$$

$$3P = P \left(1 + \frac{10R}{100} \right)$$

$$2 = \frac{R}{10}$$

$$R = 20\%$$

$$9P = P \left(1 + \frac{T \times 20}{100} \right)$$

$$8 = \frac{T}{5}$$

$$\therefore T = 40 \text{ yrs}$$

[MCQ]

4 times



#Q. An investment quadruple itself in 15 years. Then in how many years it would be ten times itself?

11 times

$\times \frac{10}{3}$
 $I = 10P$
 $A = 11P$

$\times \frac{10}{3}$
 $A = 4P \rightarrow 15 \text{ yrs}$
 $I = 3P$

$\rightarrow 50 \text{ yrs}$

$\times 3$
 $I = 9P$
 $A = 10P$

$\times 3$
 $\rightarrow 45 \text{ yrs}$

[MCQ]



#Q. An investment doubles itself in 6 years on compound interest. Then in how many years it would be sixteen times itself?

$$A = P \left(1 + \frac{r}{100} \right)^n$$
$$(2P)^4 = P \left(1 + \frac{r}{100} \right)^{6 \times 4}$$

$$2P \rightarrow 6 \text{ years} \times 1$$
$$16P \rightarrow 24 \text{ years}$$

[MCQ]



#Q. An investment triples itself in 11 years. Then in how many years it would be twenty seven times itself?

$A = 3P \rightarrow 11 \text{ years}$

$I = 2P$

$\times 13$

$I = 26P$

$A = 27P$

$\rightarrow$

143 years

$\times 13$

C.I?

$A = 3P \rightarrow 11 \text{ yrs}$

$()^3 \times 3$

$A = 27P \rightarrow 33 \text{ yrs}$

[MCQ]



#Q. A sum of $\text{₹}12000$ deposited at compound interest becomes double after 5 years. After 20 years, it will become?

$$16 \times 12,000 = 1,92,000$$

$()^4$

$2P \rightarrow 5 \text{ years}$

$\times 4$

$16P$

$\rightarrow 20 \text{ years}$

• A sum of ₹12000 deposited at compound interest becomes double after 5 years. After 20 years, it will become?

• Sol: The rate of interest at which ₹12000 doubles after 5 years is given by

$$R = 100 \left(x^{\frac{1}{5}} - 1 \right)$$

$$= 100 \left(2^{\frac{1}{5}} - 1 \right)$$

$$= 100 \times (1.1486 - 1)$$

$$= 100 \times 0.1486 = 14.86\%$$

$$A = ₹12000 \left(1 + \frac{14.86}{100} \right)^{20} = ₹12000 (1.1486)^{20}$$

$$= ₹12000 \times 15.97 = ₹191671.474$$

1,92,000

[MCQ]



#Q. What is the compound interest on ₹2500 for two years at the rate of interest 4% per annum?

Assignment

- A** ₹ 180
- B** ₹ 210
- C** ₹ 204
- D** ₹ 220

[MCQ]



#Q. A sum of money was invested at simple interest at a certain rate for 3 years. Had it been invested at a 4% higher rate, it would have fetched ₹480 more. Find the principal.

Assignment

- A** ₹ 3000
- B** ₹ 4000
- C** ₹ 5000
- D** ₹ 6000

[MCQ]

#Q. In what time a sum of money will double itself at a rate of S.I. of 6% per annum?

Assignment

- A** 6 $\frac{1}{3}$ years
- B** 16 $\frac{2}{3}$ years
- C** 33 $\frac{1}{3}$ years
- D** 21 years

[MCQ]



#Q. In how many years will a sum of ₹800 at 10% per annum compounded semi-annually become ₹926.10?

Assignment

- A** 1 $\frac{1}{3}$ years
- B** 1 $\frac{1}{2}$ years
- C** 2 $\frac{1}{3}$ years
- D** 2 $\frac{1}{2}$ years

[MCQ]



#Q. The difference between simple interest and compound interest on a sum for 2 years at 8% when the interest is compounded annually is ₹16. If the interest were compounded half yearly the difference in two interests would be nearly

Assignment

- A** ₹ 21.25
- B** ₹ 24.64
- C** ₹ 27.85
- D** ₹ 29.94

[MCQ]



#Q. Raju invested Rs.5488 for 3 years at C.I. and received an amount of Rs.6750 on maturity. What is rate percent?

Assignment

- A** $7\frac{1}{2}\%$
- B** $9\frac{2}{3}\%$
- C** $3\frac{1}{3}\%$
- D** $6\frac{1}{3}\%$



2 mins Summary



Topic

Interest



THANK - YOU